

RESEARCH EDITION · ALPHA 0.5 · 1 NOVEMBER 2026

xGEO
CORE

An open-source engine for modelling space operations: its physical models, the sources they rest on, how each is checked, and how to use it

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BUILD	TESTS	COMPARISONS	ENGINE
Alpha 0.5	64	681 / 681 pass	Godot 4.7

681

of 681 comparisons within tolerance

64

automated tests against independent references

14

physical domains, from orbits to infrared

4

known simplifications, reported, not hidden

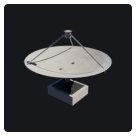
ABSTRACT

xGEO Core is an open-source engine, under the Apache License 2.0, for modelling and simulating operations in space. It computes how craft move and manoeuvre under real gravity; how radio and radar signals travel and are detected; how hot craft are and how bright they look; what sensors on the ground and in orbit can see; how an observer builds a track from what it sees; and what jamming, dazzling and other effects do. A scenario is a data file, so the engine is not tied to any one situation. Runs are reproducible: a master seed and a separate random stream for each part of the engine make run K of seed N the same world on any computer. Core runs with no window, from a command line that writes JSON lines, and plays many runs at once for Monte Carlo studies. This edition sets out each model: what it takes in, what it gives out, the equation, the source, where it stops being valid, and how sure we can be of it. Numbers chosen for play or for a scenario are marked as *design values*. The models are checked against JPL Horizons, hapsira, scipy, astropy, filterpy, ITU-R recommendations and published worked examples: 681 comparisons across 64 tests, all within tolerance, and 4 further rows that report known simplifications. Section 12 shows the models working together on one real case, a Russian inspector satellite shadowing a commercial radar satellite in October 2026.

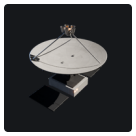
FIG. 1 WHAT CORE MODELS, AND WHERE TO READ ABOUT IT



ORBITS & MANOEUVRES
§2–3 · frames, J2, Sun and Moon, Lambert



RADIO & RADAR
§4 · ITU patterns, Friis, DF, ground radar



EFFECTS
§5 · jamming, breakup, escalation



HEAT & SIGNATURES
§6 · energy balance, IR, shadow



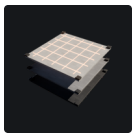
TRACKING
§7 · unscented Kalman filter, custody



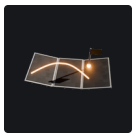
BEHAVIOUR & DOCTRINE
§8 · an opponent's rules, the kill chain



VERIFICATION
§9 · 681 comparisons, 64 tests



USING CORE
§11 · command line, seeds, Monte Carlo



WORKED EXAMPLE
§12 · Close Pass, October 2026

Every model is tested against an independent reference (§9). Click a box to jump to its section.

1 WHAT xGEO CORE IS

xGEO Core is a simulation engine for space operations. It has no graphics. It holds the physical models and the running world that ties them together. The game, xGEO Live, draws what Core computes and sends orders into it, and nothing more. Core never depends on the game, and a check in the test suite fails if it does.

Core is for teaching, concept studies and research. It is open source, under the Apache License 2.0 (§11.5). It is not a validated operational tool.

1.1 What it models

- **Orbits and manoeuvres** (§2–3). Frames and time, the Sun and the Moon, ground sites on the turning Earth, gravity with the central body's flattening and third bodies, an error-controlled integrator, Kepler and Lambert solvers, burns refined under full gravity, a planner, and flights between planets.
- **Radio and radar** (§4). Antenna patterns, link budgets, detection of unknown radio bursts, direction finding, identification, and radar on craft and on the ground.
- **Effects** (§5). Jamming, spoofing, dazzling, high-power microwaves, cyber takeover, kinetic kills and the debris they make, and what each costs in escalation.
- **Signatures** (§6). Heat balance, infrared emission, the Sun's light and the Earth's shadow, and reflected light.
- **Tracking** (§7). How an observer turns detections into a track with an uncertainty, and keeps it between looks.
- **Behaviour and doctrine** (§8). An opponent that follows rules whose settings are data, and the steps of play mapped to US joint doctrine.

1.2 Scenarios are data

A scenario is one JSON file. It names the central body (the Earth or Jupiter) and gives the craft, their classes and equipment, where each starts, the ground sites, the opponent's plan and settings, the objectives and the score. Core loads it on top of a shared file of classes and constants, `godot/data/game.json`. No scenario is built into the engine. Run with none, Core loads a sample world at Jupiter from the shared file alone. Every value in a data file is either cited, in a `sources` entry, or marked as a design value.

So this edition keeps two things apart. The models are the engine. The sizes, sites and settings a model is fed are a scenario's data. Section 12 shows one scenario, Close Pass, from start to finish. It is part of the game and is not covered by Core's licence.

1.3 Same seed, same world

Every random draw comes from one master seed. Each part of the engine draws from its own stream: setup, sensing (noise, detections, false alarms), effects, breakup, the opponent, and civil traffic. A stream's seed is the first 60 bits of the SHA-256 hash of the text `xgeo/<seed>/<run>/<name>`. So more draws in one part, a new sensor say, never move another part's numbers. The same scenario file, seed, run number, orders and snapshot step give the same output, byte for byte. Run K of seed N is the same world on any computer, and runs 0 to K − 1 make a Monte Carlo set that anyone can repeat.

1.4 Run from a command line

Core runs headless. A command-line runner loads a scenario, steps time and writes what happens as JSON lines, one object per line. Orders can be given as data too: a course of action is a JSON file of orders at set times. A batch runner plays many runs for each course of action and writes one line of outcomes per run. Because run k is the same world for every course of action, two courses of action can be compared run by run, with much less noise than independent runs give. Section 11 has the details.

1.5 Checked against independent references

A Godot script runs test cases through the engine's own functions and writes the results out. A Python suite then compares them with independent references: JPL Horizons, hapsira, scipy, astropy, filterpy, ITU-R recommendations and published worked examples. On 1 November 2026 it made 681 comparisons across 64 tests. All 681 are within tolerance, and 4 further rows report known simplifications (§9). Python never reimplements the engine in order to grade it. The engine runs in Godot 4.7, in GDScript, with 64-bit arithmetic wherever precision matters.

1.6 How this edition labels its claims

The rule is that every number is either sourced or labelled. Each source claim carries one of four labels:

- V-TEXT The document was fetched and the passage read; the section, table or page is given.
- V-RECORD The work exists (publisher, DOI or official listing) but the passage was not read, because of a paywall or because it is a book. The equation used is the standard textbook form.
- UNVERIFIED Not confirmed. Treat it as unsourced.
- DERIVED Computed from verified inputs, with the method stated so it can be rerun.

Values chosen for play, or for a scenario, carry DESIGN VALUE. A design value is never presented as measured. Where possible, it is chosen inside a range that a source does support. Most design values belong to scenarios: the sizes of craft whose real sizes are not public, the opponent's plan and its thresholds, escalation costs, objectives and the score. A few belong to the engine, such as a detector's false-alarm rate; each is marked where it is used. Section 12 lists the design values of the worked example.

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2 FRAMES, TIME AND STARTING STATES

2.1 Reference frame and time

Each world is centred on one body, its central body, which the scenario names. At the Earth, the axes are those of GCRF, the Earth-centred frame aligned with the ICRF: the Earth's mean equator and equinox of J2000

to within the frame bias of a few milliarcseconds. The scenario's central-body entry gives the Earth's shape and gravity: an oblate body of 6,378.137 × 6,356.752 km (WGS 84), with GM = 398,600.4418 $km^3\ s^{-2}$ and J2 = 1.08262668×10^{-3} (EGM2008, as tabulated by Vallado[18]). Run with no scenario, the world is centred on Jupiter, with its own J2 and the four Galilean moons. The clock is UTC, from a start time the scenario sets. Leap seconds, TT − UTC and UT1 − UTC are ignored; §2.4 gives what that costs.

2.2 Starting states

Inputs: each craft's position and velocity at the start, in kilometres and km/s, in GCRF about the central body. The engine takes this state as given. For a real satellite, the state comes from its public element set. The scenario's author propagates the set to the start time with SGP4 (Vallado et al. 2006[167]), whose output is in the TEME frame, and rotates it into GCRF. SGP4 is used only for this. From the start, every craft is integrated under the force model of §3, which is how a real analyst turns a catalogue orbit into a precise one.

A neutral example: a satellite on a near-circular orbit with semi-major axis 6,924.4 km is 549 km up, goes round in 95.57 minutes and moves at 7.584 km/s. A second craft placed a few kilometres behind it on the path that does not drift away along-track will hold its distance, which is how an inspector holds station.

An observer need not know the truth at the start. A scenario can list objects in a public catalogue; the observer then starts with a track on each, at its true state plus a random error of a size the scenario sets (§7).

2.3 The Sun and the Moon

The Sun's position comes from JPL's approximate Keplerian elements for the Earth–Moon barycentre (Standish & Williams[2], Table 1, valid 1800–2050), reversed to give the Sun as seen from the Earth: 0.9938 AU on 26 October 2026, for example. The barycentre lies about 4,700 km from the Earth's centre, which turns the Sun's direction by under 7 arcseconds. The Moon comes from the low-precision lunar series of Montenbruck & Gill[20] (§3.3.2, eqs. 3.47–3.48), good to a few arcminutes, about 500 km at the Moon's distance. GM values come from DE440[3] (the Moon's 4,902.800118 $km^3\ s^{-2}$). Planets are added only beyond the central body's sphere of influence, where their pull starts to matter.

2.4 The turning Earth and its ground sites

Ground sites, such as radars or a station that commands a satellite, sit on the turning Earth. Inputs: latitude, longitude and height on WGS 84. A site's position is rotated from the Earth-fixed frame into GCRF by Greenwich mean sidereal time (GMST 1982, Vallado[18] Alg. 15) and by IAU 1976 precession (Alg. 23), at the Earth's rate of 7.292115×10^{-5} rad/s. Nutation (under 20″), polar motion (under 0.5″) and UT1 − UTC (under 0.9 s) are omitted, so a site sits within about 1 km of where a full IERS reduction would put it. A radar measures what it sees from that point. A 1 km site error is about a thousandth of a typical range, and it is shared by every look from that site.

3 GRAVITY AND TRAJECTORY DESIGN

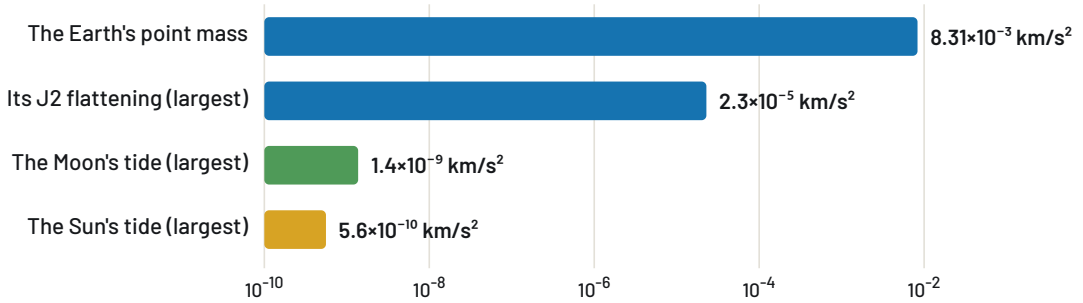
3.1 Force model and integration

The acceleration on a craft is the central body's point mass, its J2 zonal term, and the Sun and the Moon as third bodies (direct minus indirect term; Montenbruck & Gill[20] §3.3, Cowell's method):

$$\mathbf{a} = -\mu\, \mathbf{r}/|\mathbf{r}|^3 + \mathbf{a}_{J2}(\mathbf{r}) + \Sigma_k\, \mu_k \left[(\mathbf{r}_k - \mathbf{r})/|\mathbf{r}_k - \mathbf{r}|^3 - \mathbf{r}_k/|\mathbf{r}_k|^3 \right]$$

Integration is DOP853, the Dormand–Prince 8(5,3) pair with error control (Hairer, Nørsett & Wanner 1993), every step held to a relative 10^{-12}, so steps lengthen on quiet arcs and shorten near a close pass, with no step rule to tune. Against scipy's DOP853 at 10^{-13} on the same forces, the engine stays within 2.8 mm over a day, and within 0.13 m over a 20-day J2 arc. The classical fourth-order Runge–Kutta step stays in the code for tests. There is no atmospheric drag (§10). At Jupiter, the third bodies are the four Galilean moons and the Sun; the external suite of §9 checks that world against JPL Horizons.

FIG. 2 **WHAT PULLS ON A SATELLITE AT 549 KM**



The force model of §3.1 for a satellite at 549 km, with the Sun and the Moon at their distances of 26 October 2026. The Sun and the Moon are a millionth of J2, but over two days they move a craft by tens of metres, which is the scale of a close approach.

For the Earth, two checks run in Godot on every change. The orbital node of a satellite at 549 km, started from a public element set, turns 1.0395° per day, against 1.0400° from J2's first-order secular rate (Vallado eqs. 9-41/42). And a craft placed on the no-drift path beside it stays within 779 m of it over a whole orbit, which shows that the starting states and the forces agree.

3.2 Two-body tools

Kepler propagation uses universal variables (Curtis[17] Alg. 3.3–3.4), with Danby's starter near e = 1 and Vallado's starter for long hyperbolic coasts (Vallado[18] Alg. 8). Lambert's problem is solved by Izzo's method (Izzo 2015[23]), including multi-revolution transfers. Across 111 test transfers it agrees with hapsira's Izzo solver to 7.3e-15, and it reproduces Curtis Example 5.2. An earlier build used Curtis Alg. 5.2; a loss of precision in its Stumpff function as $z \to 4\pi^2$ made every transfer over 180° fail, which was one reason for the change.

3.3 Refining under full gravity

Over a 3-hour coast in low orbit, J2 alone moves a craft by kilometres, so a two-body Lambert solution misses a target only a few kilometres away. Every planned burn is therefore refined by differential-correction shooting through the full-gravity coast (Vallado[18]; single shooting as set out in NASA/TP-20220014814 §5.2.1). Newton's method starts from a two-body Jacobian and updates it with Broyden's method[24]. For example, a 3-hour move from 20 km behind a satellite at 549 km to beside it is planned at 0.74 m/s and flies at 0.75 m/s.

3.4 The planner

The planner shows a porkchop plot: departure wait against time of flight, each cell a refined transfer coloured by the Δv to go and the Δv to match on arrival. Its grid steps are design values. Close to a target, the planner also offers station-keeping: holding a point a set distance ahead, behind, above or beside the target. A point 1 km out of a target's orbital plane is held at 1.00 km.

3.5 Leaving Earth orbit

Where a scenario allows it (its `interplanetary` switch), the planner plans between planets with patched conics (Curtis ch. 8; Vallado ch. 12). A heliocentric Lambert arc runs between the planets; the escape burn is fitted so its hyperbola leaves from the craft's actual position; and the plan is refined under full gravity. Spheres of influence follow Laplace, $r = a(m/M)^{2/5}$ (Curtis §8.4). From a 549 km orbit, the cheapest departure for Mars in the 400 days after 26 October 2026 costs 4.03 km/s. That is far more than a small inspector carries (§12.2).

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4 RADIO AND RADAR: ANTENNAS, LINKS AND DETECTION

Inputs for every radio model come from the craft's class in the scenario: transmit power, dish size, frequency, bandwidth, burst length and schedule for an emitter; dish size, system noise temperature and

interferometer baseline for a listener. The models below turn those into a received power, a chance of detection and a bearing with its error.

4.1 Antenna gain and patterns

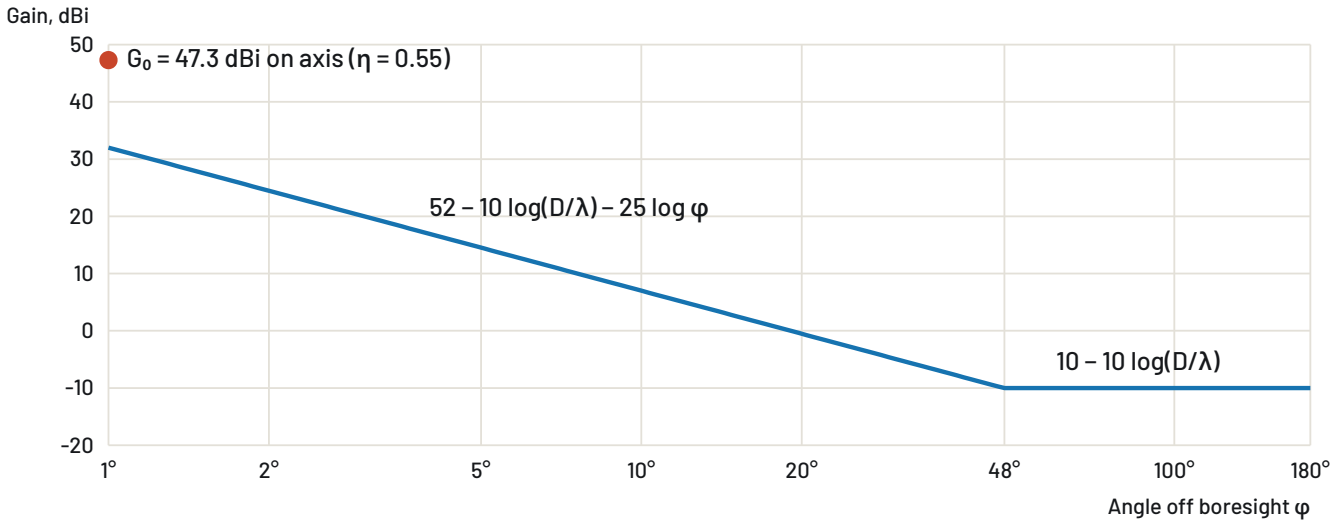
The peak gain of a dish of diameter D at wavelength λ is $G_0 = \eta(\pi D/\lambda)^2$, with $\eta = 0.55$ throughout. That is 1.24 dB conservative against the DSN 34 m antenna at X band (68.24 dBi at 8425 MHz, DSN 810-005 Module 104 Rev. O Table 8[39]). The beamwidth is about 70λ/D degrees (NAWCWD TP 8347 §3-1[54]; ITU-R SA.509-3 §1.3[33]) V-TEXT.

Off-axis gain follows the ITU reference envelopes: SA.509-3 for large dishes ($D/\lambda \geq 100$), and S.465-5 Note 4 for small ones[34]:

```
G(ϕ) = 52 − 10 log(D/λ) − 25 log ϕ   dBi,   100λ/D ≤ ϕ < 48°
G(ϕ) = 10 − 10 log(D/λ)              dBi,   48° ≤ ϕ ≤ 180°
```

These are envelopes of sidelobe peaks, used in interference coordination, so they are conservative for a well-designed antenna. Note 2 of S.465 warns against using them below $D/\lambda = 50$, and the small dishes on small satellites are often that small. They are still the most credible public pattern. Every emitter points: a command radio aims its dish at its ground station, so a listener off that line hears only a sidelobe. Ships and buoys can also listen through four-face arrays that form beams in all directions (after the TDRS S-band multiple-access array, NASA TM-105422), with gain equal to the face's peak plus 10 log cos(scan).

FIG. 3 REFERENCE ANTENNA ENVELOPE



ITU-R SA.509-3 sidelobe envelope drawn for an illustrative dish 100 wavelengths across. It is an envelope of sidelobe peaks, so it is conservative for a well-designed antenna.

4.2 Link budget and noise

$$P_r = P_t\, G_t(\theta_e)\, G_r(\theta_r)\, (\lambda / 4\pi R)^2 \qquad \text{(Friis 1946)}$$

$$N = k\, T_s\, B$$

Free-space loss is the exact $20 \log(4\pi R/\lambda)$ of ITU-R P.525-4 eq. (3). The practical eq. (4) rounds its constant to 32.4 dB (exactly 32.448 dB), so the engine differs from eq. (4) by 0.048 dB. The model is free space only: it leaves out the atmosphere, plasma and multipath (§10).

4.3 Detection of unknown bursts

Listeners don't know a burst's waveform, so they detect it with an energy (radiometer) detector (Urkowitz 1967[51]). The integrated SNR grows as $\sqrt{(B\tau)}$. The chance of detection is:

$$P_d = Q\left(\left(Q^{-1}(P_{fa}) - d\right) / \sqrt{(1 + 2s)} \right), \qquad P_{fa} = 10^{-6}$$

The false-alarm rate is a **DESIGN VALUE**. This expression matches the exact chi-square distributions to 0.01 for $n \geq 10^5$ samples, and even odds come at 6.8 dB. Below that many samples the Gaussian form is less exact.

Spreading a signal over a wider band costs an eavesdropper's energy detector $\sqrt{(B_{old}/B_{new})}$ in SNR at fixed power (the radiometer equation; Urkowitz 1967[51]). ITU-R SM.1055[38] §1 notes that spread signals are noise-like and not readily detected.

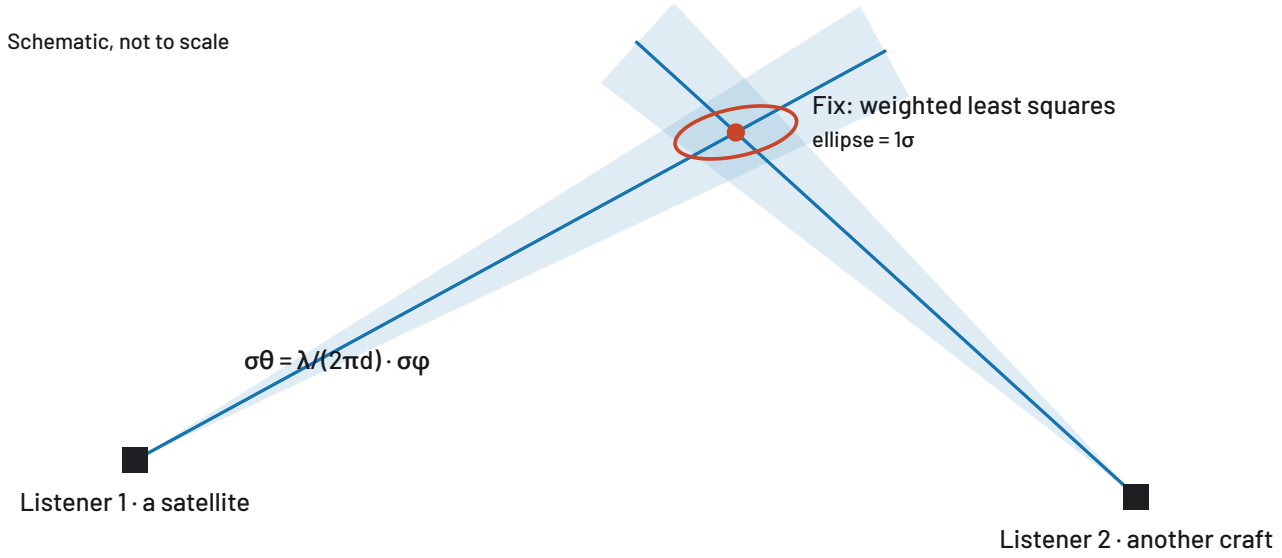
4.4 Direction finding and identification

A two-element interferometer measures a bearing with

$$\sigma_\theta = \lambda/(2\pi d)\cdot\sigma_\phi, \qquad \sigma_\phi^2 = (1+2s)/(2B\tau\, s^2)$$

This is the cross-correlation phase error for a noise-like signal. It follows from the Knapp & Carter (1976) Cramér–Rao bound in terms of coherence, $\gamma^2 = s^2/(1+s)^2$, in the narrowband limit; it is checked by a Monte Carlo of a two-element correlator to 3%. A 20 µrad floor is a DESIGN VALUE. Bearings from separated listeners are crossed by weighted least squares. Identification compares a recording's modulation and symbol rate against a threat library above a minimum SNR, a design-value gate. Specific-emitter identification is real: it recognises a device by imperfections in its analog circuitry (Danev et al.[78]).

FIG. 4 DIRECTION FINDING



Two listeners (two craft, or two receivers on one craft a few metres apart) each take a bearing with its own error wedge. The fix is their weighted least-squares crossing; the farther apart the listeners, the tighter the ellipse. Schematic.

4.5 Ground radar

A ground radar is a site on the turning Earth (§2.4) whose class carries a radar block. Inputs, all scenario data: range noise σ, angle noise σ, the elevation mask and the reach. Output: range, azimuth and elevation, each with Gaussian noise of the stated size. Where an operator has published its radar's accuracy, the site uses those figures and sees whatever is above its mask and inside its reach, with no signal-to-noise test. Otherwise the radar equation decides (SNR 20; Skolnik), with monopulse angle noise. A site measures every 10 s (the cadence in its radar block) all through a pass, between the 600 s looks too. To know when to stop for those fixes, the engine finds the minutes when anything could be in a site's view on Kepler orbits, once a minute with a margin, then decides each fix on the true state at its moment. A site moves with the Earth: at 65.1° N it moves 704 km in an hour, at 0.1962 km/s; the Earth's rate times the site's 2,690 km distance from the polar axis on WGS 84 gives 0.1961 km/s (DERIVED).

A radar on the ground differs from one on a craft in two ways that matter. It sees a satellite in low orbit only for a few minutes per pass, so between passes the track rests on the filter's prediction (§7). And its range noise, tens of metres, is far smaller than its angle noise at 1,000 km (0.05° is 870 m across the line of sight), so a single pass fixes how far away a craft is much better than where across the sky. Section 12.3 gives five real sites as a scenario uses them.

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5 EFFECTS AND ESCALATION

The engine models jamming, spoofing, dazzling, high-power microwaves, cyber takeover and kinetic kills. A scenario turns them on by giving a block of settings: every power, threshold and escalation cost below is scenario data. The values quoted are the design values of the engine's own test scenarios. The equations are the engine's.

5.1 The jamming equation

Jamming a command uplink means being louder, at the victim's receiver, than its ground station. Both paths are one-way, so the equation is Friis applied twice. This is the communications form of NAWCWD TP 8347 §4-9 V-TEXT, and Adamy's EW 101[75] gives it in dB form:

$$J = P_j + G_j(\text{aim error}) + G_v(\theta_j) - L_j$$

$$S = P_u + G_u + G_v(0) - L_s$$

$$J/S = J - S - 10\log(B_j/B_v) \quad \text{dB}$$

$G_v(\theta_j)$ is the victim's gain toward the jammer, at angle θ_j off its line to its ground station, so geometry matters: a jammer near that line faces the victim's main beam. The aim term means a loose track costs beam-pointing gain. Before the signal is identified, a jammer must spread its power across the whole allocation (barrage);

once identified, it can put it all in the channel (spot). How much J/S breaks a link has no universal number. NAWCWD §4-7 (p. 4-7.3) says plainly that effectiveness "is not a direct mathematical function of J/S" V-TEXT. The test scenarios require +6 dB, a DESIGN VALUE inside the 0–10 dB band typical for FEC-coded BPSK. The effect is all or nothing: the link is cut or it is not, and the victim does not hop frequency or steer a null.

5.2 Other effects

EFFECT	MODEL	SOURCE / STATUS
Spoof (FAKE)	False ranging tones capture the target's ranging loop at J/S ≥ 0 dB, inducing a drift	Loop capture at 0 dB and a drift of 900 km/h are design values
Dazzle	Spot radius $R\sqrt{((1.22\lambda/D)^2 + j^2)}$ at 1.06 µm, 2 m aperture, 0.2 µrad jitter; 10 W/m² at the aperture	Airy disc $1.22\lambda/D$ (Hecht); jitter added in quadrature (engineering approximation); 10 W/m² is an engineering estimate allowing for an OD-6 filter
HPM (PULSE)	Far field $E = \sqrt{(30PG)}/R$ from a 1 GW-peak X-band source; front-door damage at 100 V/m	$E = \sqrt{(30PG)}/R$ (Balanis; ITU-R P.525-4 eq. 1); 100 V/m is a DESIGN VALUE (engineering estimate); GW-class backward-wave oscillators (Benford et al. 2016)
Cyber (TAKE)	Takeover after recording enough command traffic	Abstracted. Real command links are encrypted, and this is stated as fiction
Kinetic kill	$P_k = \text{rel} \cdot (1 - e^{-r^2/2\sigma^2})$, from divert reach against track uncertainty	Rayleigh. Breakup by the NASA standard breakup model (below). The 1,200 catalogued pieces are a DESIGN VALUE set below FY-1C (2007; 3,442 catalogued[138]) and Cosmos 1408 (2021)

A kill breaks the wreck up by the NASA standard breakup model (Johnson et al. 2001[131], as given in Frey & Colombo 2021 App. B[135]). An impact is catastrophic above 40 J/g, and $N(\geq L_c) = 0.1\ M^{0.75}\ L_c^{-1.71}$ fragments are drawn. Each fragment's area-to-mass ratio $\chi = \log_{10}(A/M)$ follows App. B: the bimodal spacecraft law above 11 cm, the small-object law below 8 cm, and a linear blend in $\log L_c$ between 8 and 11 cm. Its speed follows $\log_{10} \Delta v \sim N(0.9\chi + 2.9, 0.4)$, Δv in m/s. The model gives the cloud at the moment of breakup; how the debris spreads over the years after is not modelled.

5.3 Escalation

Each effect carries an escalation cost. In the test scenarios: jamming 2 per hour, radar 1 per hour, spoofing 4, dazzling 8, cyber 10, high-power microwave 15, and a kinetic kill 45. All are design values. Doctrine provides no escalation ladder (§8.2).

A scenario can also price acts that are not effects at all, such as an approach. These tension rules are data too: a cost when the opponent parks too close to the craft it is watching, a cost each time one side crowds the other into giving way, a cost per hour spent too close, the levels at which the opponent turns cautious or leaves, and the level that counts as open conflict. Section 12.5 shows one set.

6 HEAT, INFRARED AND VISIBLE SIGNATURES

Inputs from each craft's class: dry mass, surface area, temperature, absorptance and emittance of its skin, any radiators, and the hot parts of its engine. Outputs: the craft's temperature over time, its infrared intensity in the sensor's band, its brightness in visible light, and the range at which a given telescope detects it.

6.1 Energy balance

Each craft has a skin and, if it carries a reactor, radiators. Skin temperature follows a single-node balance with heat capacity (Gilmore[103]):

$$C\,dT/dt = P_{leak} + \alpha\,S\,f\,A/4 - \varepsilon\,\sigma\,T^4\,A, \qquad C = 897\ \mathrm{J\,kg^{-1}\,K^{-1}} \times 0.25 \times m_{dry}$$

$S = 1{,}361\ \mathrm{W\,m^{-2}}$ at 1 AU (Kopp & Lean 2011[109]), scaled by the inverse square of the Sun's distance, and f is the sunlit share (§6.4). The heat capacity is aluminium's at 298 K, 24.21 J $\mathrm{mol^{-1}\,K^{-1}}$ or 897 J $\mathrm{kg^{-1}\,K^{-1}}$ (NIST-JANAF, NIST Chemistry WebBook[105]); Incropera[104] Table A.1 gives 903 J $\mathrm{kg^{-1}\,K^{-1}}$ at 300 K, 0.7% more. The 25% skin fraction is a DESIGN VALUE. A craft of 600 kg dry mass has a skin time constant of about 1.5 hours (DERIVED: $C = 897 \times 0.25 \times 600\ \mathrm{kg} = 134{,}550\ \mathrm{J/K}$ against $4\varepsilon\sigma T^3 A \approx 25\ \mathrm{W/K}$). So it cools noticeably during each pass through the Earth's shadow, which lasts up to about 36 minutes at 549 km. The Earth's own reflected sunlight (albedo 0.367) and its infrared glow (254 K as a black body) are not yet added to the heat balance (§10). Absorptance and emittance come from NASA RP-1121 (Henninger 1984[95]).

The engine also models reactor radiators, after the 100 kWe Brayton plant of NASA Glenn's JIMO study (Mason 2003[96]; Siamidis 2005[98]), as two-sided Lambertian plates, $I(\theta) = \varepsilon\sigma T^4 A \cos\theta/\pi$.

6.2 Going dark

A craft with a reactor can stop radiating for a while by storing its heat in water it vents (2.5 MJ/kg) or ice it melts (333 kJ/kg, Leimkuehler 2011[102]). The time it can stay dark is the stored mass times that energy, divided by the heat it must hide.

6.3 In-band infrared detection

Telescopes work in NEO Surveyor's NC2 band, 6–10 µm (Mainzer et al. 2023[115] V-TEXT). The in-band share of each body's emission comes from the Widger & Woodall series, which matches astropy to 6×10^{-11}. The 0.5 m

LWIR's 1σ noise-equivalent irradiance is 7×10^{-18} W m^{-2}. That lies inside the 0.4–1.1×10^{-17} range derived from NEO Surveyor's S/N-5 sensitivity. A look detects with

$$S/N = I / R^2 / (NEI \times background), \qquad P_d = \Phi(S/N - 5)$$

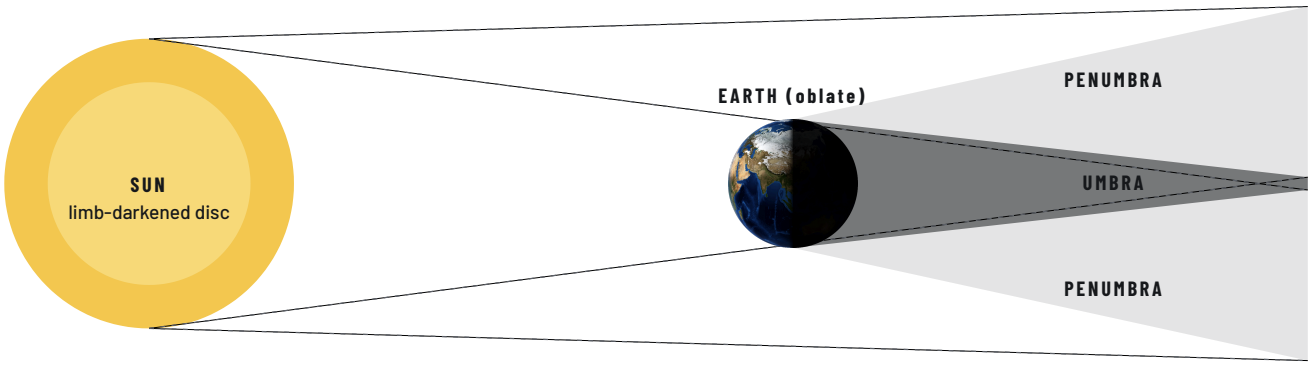
Against the planet's disc, the background raises the NEI ×50, a DESIGN VALUE inside the ×26–92 derived for a background-limited sensor. Just off the disc, stray light raises it ×5, also a design value, out to three times the planet's apparent radius but never more than 20° past its limb. From low orbit the Earth fills almost half the sky, and three times its apparent radius would leave no sky at all; 20° is about the Hubble Space Telescope's bright-Earth avoidance angle. False alarms are Poisson, with a mean of 0.1 per sensor per look (design value), and only lines from two places at once start a track.

6.4 Shadows and reflected light

The Sun is a limb-darkened disc, $I(\mu) = 1 - u(1 - \mu)$ with $u = 0.6$ (Allen's[106] §14.7), sampled by 397 rays in 12 rings. The sunlit share is the weighted fraction not blocked by the oblate Earth (6,378 × 6,357 km), the Moon, or, far out, a planet. This gives both umbra and penumbra (Montenbruck & Gill §3.4.2), within 0.019 of a dense numerical integration.

Sunlit hulls are also seen in visible light. The Sun's $M_V = 4.81$ (Willmer 2018) is scaled to the irradiance at the craft, and each hull is a Lambert sphere with phase law $\Phi(\alpha) = (\sin\alpha + (\pi - \alpha)\cos\alpha)/\pi$. It returns a share $p\cdot\Phi(\alpha)\cdot r^2/R^2$ of the sunlight at its distance R, with p the geometric albedo and $r^2 = A/4\pi$ for a hull of surface area A (Madhusudhan & Burrows 2012 eq. 3[122]). An earlier build used the cross-section πr^2 in place of r^2, which made every hull π times (1.24 mag) too bright and every visible range $\sqrt{\pi} = 1.77$ times too long. A 1 m GEODSS-class telescope reaches magnitude 22 (Stokes et al. 1996[119]; the free 1998 Lincoln Laboratory Journal account of the same tests gives magnitude 22 at S/N 4 in under 100 s). The engine asks for S/N 5 at magnitude 22, a DESIGN VALUE. Visible telescopes look only where a track already exists, because reaching magnitude 22 takes about 100 s staring at one field. A real hull is not a sphere: glints from flat panels can make it much brighter for a moment, and the model does not show them.

FIG. 5 **UMBRA AND PENUMBRA**



Schematic, not to scale. 397 rays in 12 rings sample the disc; the share not blocked is the sunlit fraction f.

Because the Sun is a disc, the Earth casts a full shadow inside a soft-edged one (§6.4). At 549 km a satellite spends up to about 36 minutes of each 95.6-minute orbit in it, depending on the season and its orbit's angle to the Sun.

6.5 What sets detection range

Two rules follow from the equations. Infrared range grows as the square root of a target's in-band intensity, so a craft burning its engine is seen about 2.4 times farther than one coasting, and seen against the Earth's disc, $\sqrt{50} \approx 7$ times nearer than against deep sky. Visible range grows with the telescope's aperture: limiting flux scales with aperture and flux falls as the square of range, so a 1 m telescope sees the same hull $\sqrt{(1/0.3)} = 1.83$ times as far as a 0.3 m one. In low Earth orbit these ranges are tens of thousands of kilometres or more, far beyond the distances of an inspection. What limits seeing there is geometry: sunlight, the Earth in the way, and when a ground site's pass comes. Section 12.4 gives the ranges for real craft.

7 TRACKING

Every contact is a track: an estimate with a covariance, not ground truth. Inputs: bearings from telescopes and receivers, and range, azimuth and elevation from radars (§4.5). Output: a position and velocity with a 1σ ellipse, carried forward in time. Away from Jupiter, tracks are three-dimensional. Measurements update a track through an unscented Kalman filter (Julier sigma points, $\kappa = 0$, circular mean for angles), which matches filterpy to 3×10^{-12}. Prediction uses central-difference state transition matrices and white-noise acceleration (Bar-Shalom et al. 2001). Two cases needed special handling.

- **Close in.** With an observer a few kilometres from its target, sigma points can straddle the sensor. Within a tenth of the range, a bearing therefore enters as a pseudo-linear measurement (Lindgren & Gong 1978; Aidala & Nardone 1982). Between craft in nearly the same orbit, range is unobservable from angles alone without a manoeuvre (Woffinden & Geller 2009). So without a recent radar range, the track keeps at least 0.5 × range of 1σ along the line of sight (design value).
- **Range after a burn.** A telescope ranges a hull of known size by the angle it spans: metres at a few km, about 1.5 km at 30 km. After a burn, a linearised filter goes on claiming the range it had before, though its velocity, and with it its range, is now off. So from a track's first burn on, the track claims no better along the line of sight than all the ranges measured since the last burn, taken together ($1/\sqrt{\Sigma 1/\sigma^2}$). One radar fix settles it; a far telescope's ranges settle it slowly.
- **Unseen burns.** A bearing whose normalised innovation passes the $\chi^2(1)$ 99.9% point (10.83) is taken as a burn nobody saw, and widens the track (Bar-Shalom ch. 11). The 50 m/s and 10 min used are design values.

Between measurements the track coasts on the force model of §3, so how good a track stays depends on how well the filter predicts. Tracks also size the burns they see: a jump in a track's motion is reported as a burn of about so many m/s, give or take its uncertainty.

A scenario can score *custody*: the share of the time a track is good. The engine counts a track as good, at every look, when both hold: its own 1σ ellipse is smaller than a distance the scenario sets (what an operator can see), and the estimate really is within that distance of the truth (what only the referee knows). So the score rewards a track that is right and knows it, not one that only claims to be. With truth reported, an analyst can compare the two over a run (§12.6).

A track nobody has detected for 24 h, or whose 1σ exceeds 100,000 km, is dropped (after Blackman & Popoli 1999). Both thresholds are design values. Measurements are matched to tracks by the true identity of the object, not by a data-association algorithm (§10). Contacts are drawn after MIL-STD-2525D, with a dashed frame for a last known position.

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8 BEHAVIOUR AND DOCTRINE

8.1 An opponent that follows rules

The opponent is not a planner that thinks ahead. It follows a short set of rules, checked at every look, whose settings are scenario data. The main rule set is the shadow: an uncooperative rendezvous and inspection of a target. Its inputs are:

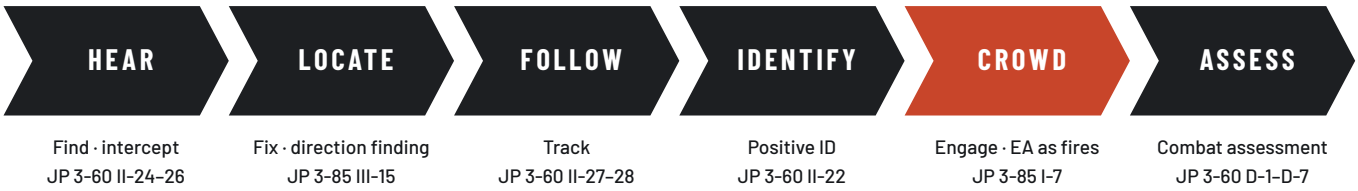
- **A plan.** A list of slots beside the target, each given as a distance along the target's track (ahead or behind), how long to hold there, and how long to take to get there. The opponent flies to each slot with two burns, which its ground station must command, holds, and moves on. After the last hold, the inspection is complete and it leaves.

- **Reactions.** With no rival craft within a set distance, it takes its next slot closer in, down to a floor. With a rival within a smaller set distance, it gives way to a slot on the far side of the target. After a set number of give-ways, it leaves with a set burn. Cut off from its ground station for a set time, with a rival nearby, it leaves; cut off but alone, it holds and waits. If a move would cost more than about 45% of its remaining Δv, it ends the inspection there.
- **Its radio.** It reports home in bursts on a schedule and after each move. Each burst is a real emission that the radio models of §4 can hear, locate and identify.

The rule that a craft cut off from its ground station backs away is grounded in practice, not in a published rule: a Progress approach to the ISS was aborted on loss of its telemetry link in 2010[162], and ESA's ATV carried RETREAT and ESCAPE modes for the same reason. Every threshold is a design value. A batch run can change any of them, or draw them at random across runs, to spread what is not known about an opponent (§11.3). The opponent does not adapt: it does not hop frequency, steer nulls or switch to optical links.

8.2 Doctrine and the kill chain

FIG. 6 THE STEPS OF PLAY AS A KILL CHAIN



Each step mapped to its doctrinal term and citation (table below).

The steps of play follow US joint doctrine, read from the published texts: JP 3-85 *Joint Electromagnetic Spectrum Operations* (2020)[139], JP 3-60 *Joint Targeting* (2018)[141], AFDP 3-60 (2026)[142] and the USSF *Space Warfighting* framework (2025)[143].

STEP	DOCTRINAL TERM	CITATION
Hear	F2T2EA *Find*; electromagnetic support search and intercept	JP 3-60 II-24–26; JP 3-85 I-6
Locate	*Fix*; direction finding, precision geolocation	JP 3-60 II-26–27; JP 3-85 III-15, B-4
Follow	*Track*	JP 3-60 II-27–28
Identify	Positive identification by EW support systems or visual recognition (doctrine acquires it during *Fix*)	JP 3-60 II-22
Crowd	No doctrinal term: a show of presence by an approach that stops short of contact. Where a scenario allows effects, this step is *Jam*, doctrine's *Engage* by electromagnetic attack (JP 3-85 I-7, I-13, GL-6, GL-7)	—
Assess	Combat assessment, with confidence graded confirmed, probable or possible	JP 3-60 D-1–D-7

Under JP 3-85, a craft going dark is practising electromagnetic protection: "multispectral low-observability" is one of its listed examples. Doctrine says nothing about TDOA/FDOA, look-through, own-platform jammer isolation, or an escalation ladder. JP 3-85, JP 3-60 and AFDP 3-60 never use the words "spoofing" or "reversible": their terms are intrusion and deception, and temporary or nonlethal effects. (The USSF framework does speak of reversible and nonreversible actions, pp. 5 and 9.) Nor does doctrine treat crowding another craft as an act at all. Escalation costs and tension rules (§5.3) are therefore entirely design values. Their ordering (reversible and deniable acts cheap, debris-generating acts ruinous) follows doctrine's escalation-of-force guidance qualitatively.

9 VERIFICATION

A Godot script runs every test case through the engine's own functions and writes the results out. A Python suite (pytest) then compares them with independent references, each comparison carrying its own tolerance. JPL data is cached, so the suite runs offline. The suite exercises the engine on its sample world at Jupiter, where JPL Horizons gives an exact reference for every body; the same models run at the Earth. On 1 November 2026 it made 681 comparisons across 64 tests; all 681 fall within tolerance, and 4 further rows report known simplifications.

MODEL	REFERENCE	TOLERANCE	WORST RESULT
Kepler propagation (94 cases, e to 0.999 and hyperbolic)	hapsira farnocchia; scipy DOP853	1e-8	7.1e-10
Lambert (111 transfers)	hapsira izzo	1e-6	7.3e-15
Worked examples	Curtis Ex. 3.7, 5.2; Vallado Ex. 2-4	½ last digit	pass
Force model (J2, third bodies)	hapsira perturbations; gradient of the J2 potential	1e-9	8.5e-11
DOP853 integration, 1 day	scipy DOP853, rtol 1e-13	1 cm	2.8 mm
J2 secular rates	Vallado eqs. 9-41/42	1%	0.40%
Galilean moons, Sun and planets	JPL Horizons	per body	pass
Antenna patterns (468 cases)	ITU-R SA.509-3, S.465-5, coded from the text	1e-9 dB	1.4e-14
Free-space loss	ITU-R P.525-4 eq. (4), 32.4 dB constant	0.05 dB	0.048 dB
Radiometer gain $\sqrt{(B\tau)}$	Monte Carlo energy detector	6%	1.5%
Interferometer DF σ	Monte Carlo two-element correlator	8%	3%
Radar SNR and monopulse angle	Skolnik; Barton	1e-9; 0.2%	5.6e-15; 0.14%
Jammer J/S and reach	Adamy, independent dB form; brentq	1e-9 dB	5.7e-14
Radiators against JIMO	Siamidis 2005 design point	2%	1.1%
Shadow fraction	Dense numpy disc integration	0.02	0.019
Skin heat capacity	scipy solve_ivp	0.05 K	0.0001 K
Unscented update	filterpy UKF	1e-6	3.3e-12
Rocket equation	Tsiolkovsky, every ship	0.006 km/s	0.0032 km/s

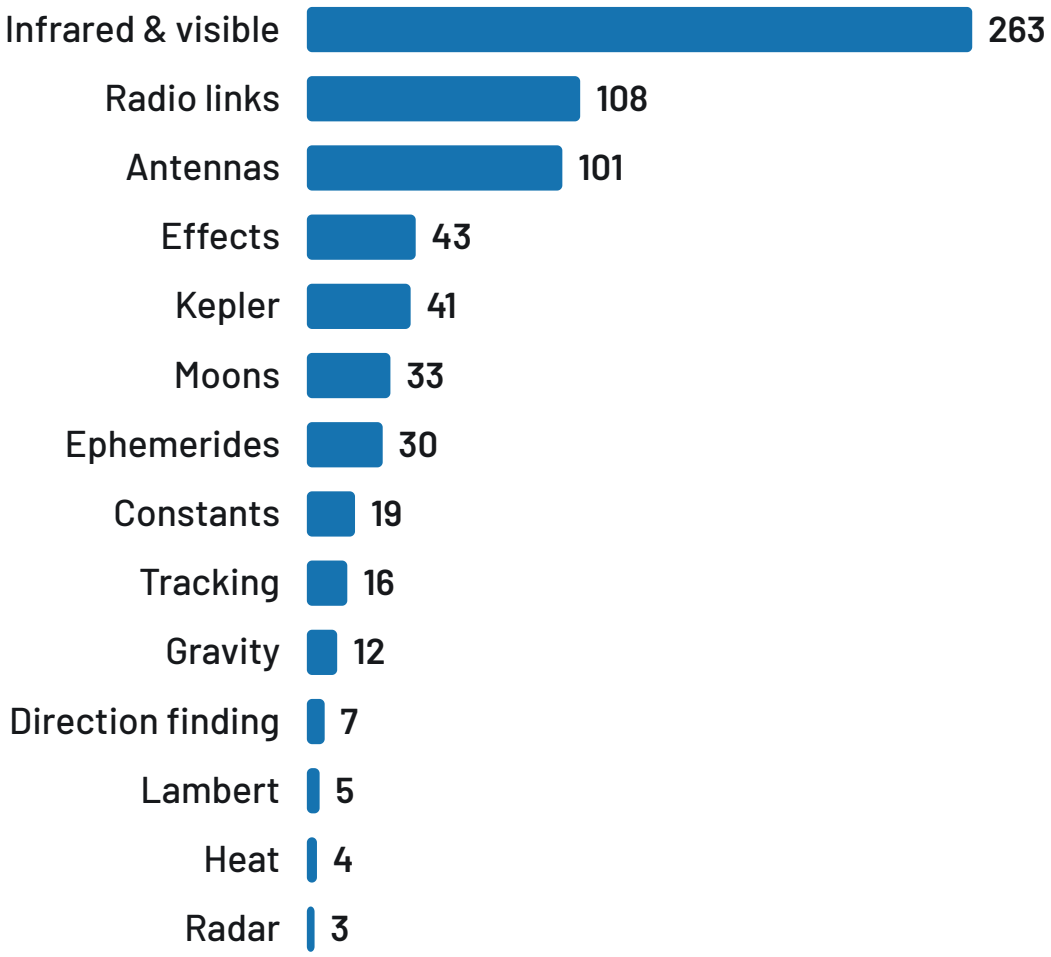
Table 1. Selected comparisons. The full table has one row per comparison.

The Earth's own numbers are checked by two Godot tests, `earth_test` and `close_pass_earth_test`, against textbook values. Both pass. The craft and sites they use are those of the worked example (§12).

CHECK	ENGINE	REFERENCE
Node of a satellite at 549 km, turn per day	1.0395°	1.0400°, J2's first-order secular rate (Vallado eqs. 9-41/42)
A craft on the no-drift path beside it, over one orbit	within 779 m	under 800 m: starting states and forces agree
The Sun's distance on 26 October 2026	0.9938 AU	0.98–1.00 AU in late October
A ground site's speed at 65.1° N	0.1962 km/s	0.1961 km/s: ω × distance from the axis at 65.1° N on WGS 84
A station 1 km out of the satellite's plane	held at 1.00 km	1 km
A 3 h move from 20 km behind to beside it	0.75 m/s flown	0.74 m/s planned
Δv of a 675 kg craft carrying 75 kg of hydrazine	259 m/s	Tsiolkovsky: 2.2 km/s × ln(675/600)

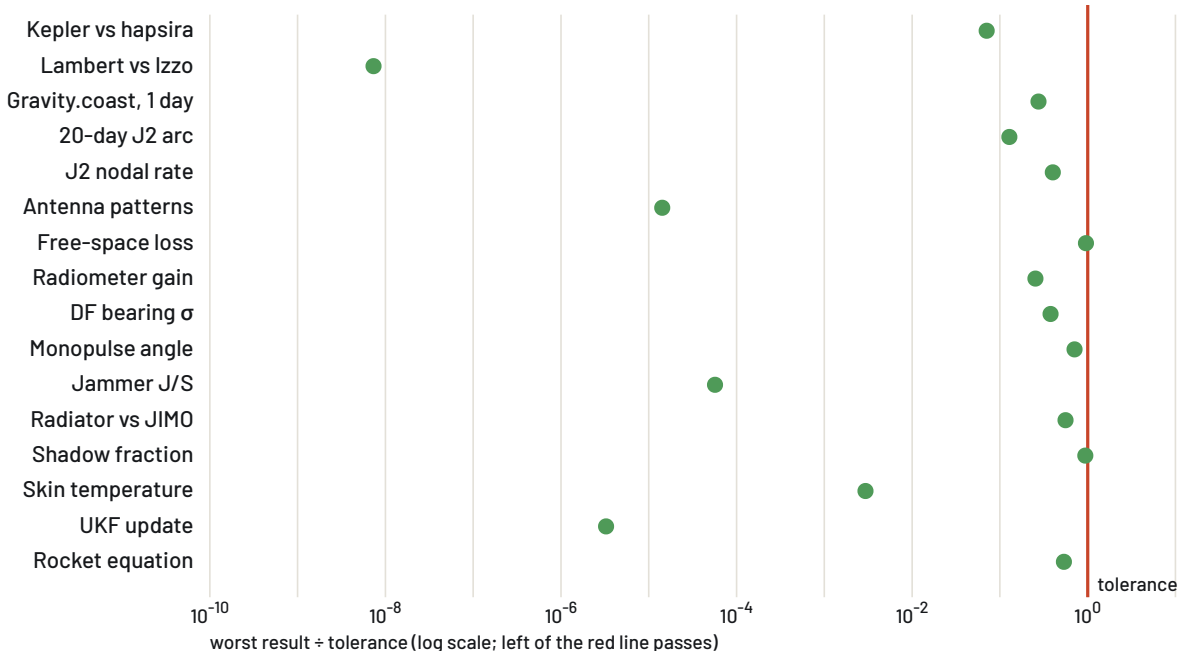
Table 2. Checks at the Earth.

FIG. 7 **WHERE THE 685 ROWS GO**



Rows per domain on 1 November 2026: 681 within tolerance, and 4 that report known simplifications.

FIG. 8 **HOW MUCH MARGIN EACH CHECK HAS**



Worst result divided by its tolerance, for the numeric rows of Table 1. Every dot sits left of the red line; most sit orders of magnitude inside it.

The suite has also found and fixed real errors. Moon longitudes were up to 114° off. DF σ was three times too large, because it was fed the integrated SNR. The barrage band was twice as wide as the real allocation. Long-way Lambert transfers failed. Two hulls were π times too bright in visible light. And the check of who sees whom had to learn the 20° cap on glare that the Earth brought in (§6.3). Kepler's equation lost accuracy near e = 1. Each fix and its effect is recorded in the project's verification log.

10 LIMITS, SIMPLIFICATIONS AND OPEN QUESTIONS

Simplifications

- No atmospheric drag. At 549 km it slows a small satellite by metres per second per year, depending on the Sun's activity. Over a run of days it moves craft relative to each other by an amount that depends on their area-to-mass ratios, which are often not public. The lower the orbit, the sooner drag matters.
- No reflected sunlight or infrared from the Earth in the heat balance, though the Earth's disc does raise the infrared background (§6.3).
- A low-precision Moon (a few arcminutes) and an approximate Sun; no gravity terms of the central body beyond J2; no solar radiation pressure or relativity.

- No nutation, polar motion or UT1 − UTC: ground sites sit within about 1 km of their true places.
- SGP4 is used only to make starting states; nothing is fitted back to element sets during a run.
- Sensors and the opponent act once per look, every 600 s. Ground radars are the exception: they measure every 10 s through each pass (§4). The 10 s cadence is a design value; real radars that keep an object in track measure more often.
- Measurements are matched to tracks by the object's true identity, so the filter never confuses two objects.
- Radio is free space only: no atmosphere, plasma or multipath. Jamming is all or nothing.
- The umpire's view: some kill-chain indicators still read ground truth rather than Blue's assessed belief.

Not verified

- HawkEye 360 geolocation accuracy; the Albersheim validity limits; specific-emitter identification accuracy; look-through duty cycles.
- The 2024 edition of JP 3-60 (the 2018 edition was read), and JP 5-0's definitions of MOE and MOP.
- MIL-STD-2525D was read through a reproduction of its tables, not the original PDF.

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11 USING xGEO CORE AS AN OPEN-SOURCE TOOL

The models in this edition make up **xGEO Core**. **xGEO Live**, the game, draws what Core computes and sends orders into it. Core runs with no window, and it is being prepared for release as an open-source simulation tool for teaching, concept studies and research. It is not a validated operational tool.

11.1 What is Core and what is Live

PART	WHAT IT HOLDS	WHERE
Core: physical models	Two-body motion, Lambert, gravity with J2 and third bodies, ephemerides, starting states, the turning Earth and ground sites, link budgets, antenna patterns, radar, heat and infrared, the heat forecast, sensing, the tracking filter, breakup and debris, civil traffic, and the world that ties them together	`godot/core/`
Core: rules of play	Effects and their costs, the opponent's behaviour, objectives and the debrief, the log's wording, save and continue, courses of action and studies. They run headless, but they are rules of play, and most of their values are Notional	`godot/core/`
Core: runners	The command-line runner and the batch runner	`godot/cli/`
Data	Shared classes, equipment and constants (`game.json`); scenario files with their craft, sites, the opponent's plan and the objectives	`godot/data/`
Live	The console, the map, the windows, art, sound, the tutorial	`godot/ui/`, `godot/scenes/`, `godot/assets/`

Table 3. The split. A check in the test suite fails if anything in Core names one of Live's classes.

11.2 Running Core without the game

A command-line runner loads a scenario, steps time and writes what happens as JSON lines, one object per line:

```
godot --headless --path godot --script res://cli/run.gd -- \
      --hours=24 --step=600 --seed=4417 --out=run.jsonl
```

The first line describes the run (scenario, seed, run number, start time, units). Each step line carries the time, escalation, Blue's craft (position, velocity and Δv left), Blue's tracks with their 1σ ellipses, and every log entry since the last step. The last line is the debrief. Tracks report Blue's belief, never the truth; `--truth=true` adds the true states for analysis. Each track in a step line is carried forward to the line's own time, so belief and truth can be compared at the same moment. The world acts at every 600 s look whatever the step, so a shorter `--step` gives finer snapshots of the same run. Units are kilometres, km/s and seconds, centred on the scenario's central body. `--scenario=` picks the scenario file; left empty, Core runs its sample world at Jupiter. The test suite runs the runner on every change.

11.3 Many runs, courses of action and studies

A course of action is a JSON file of orders at set times: burns, intercepts, matching a target's velocity, effects, buoys, analysis of what is heard, and radar on or off. Each order goes through the same functions a player uses, so it plays by the same rules. A transfer to an opponent's craft is planned on Blue's track of it, not on its true state.

The batch runner, `cli/batch.gd`, plays runs 0 to N − 1 of a seed for each course of action, and writes one line of outcomes per run. Several processes can share a batch by taking a range of runs each. Run k is the same world for every course of action: the same opening track error, the same detections and the same chance draws until the orders change what happens. These are common random numbers (Law 2015, *Simulation Modeling and Analysis*, 5th ed., §11.2), and they let two courses of action be compared run by run with much less noise. Any number in a scenario can be changed for a batch without editing the file: fixed for every run (`--set`), swept over a list of values (`--vary`), or drawn afresh for each run between bounds (`--sample`), which spreads what is not known over the runs.

A study runs a batch across all the computer's cores and summarises it: for each course of action, the share of runs won and the mean score, fuel, custody and closest approach, each with its 95% interval; the paired difference from the first course of action; and which sampled settings drive the score. Shares carry Wilson score intervals (Wilson 1927); means and paired differences carry Student's t intervals (Law 2015, §9.4 and §10.2); drivers are ranked by Spearman's rank correlation. A study answers "which course of action does better in this model", not "what would happen": the score, the objectives and the opponent's behaviour are a scenario's design values.

11.4 Real, Projected and Notional

Every model and value carries one of three labels, which extend the source labels of §1.6:

- **Real:** sourced and checked against an independent reference (§9). The orbits, gravity, the radio and radar equations, Planck's law, the filter.
- **Projected:** near-future and grounded in programmes that exist today, such as nuclear-thermal drives at NASA's tested 900 s specific impulse.
- **Notional:** invented for a scenario, the way defence and industry studies fly notional systems: an opponent's classes and behaviour, equipment sizes, escalation scores, a compressed timescale.

A Real model fed Notional inputs gives a Notional answer: the jamming equation is exact, but a jammer's power in a scenario is a design value, and so is the range it yields. The project's fidelity table (`docs/FIDELITY.md`) gives each model's label, what it was checked against and what it leaves out.

11.5 Status of the release

Licence (26 October 2026): xGEO Core, the code of the models, the runners, the tests and the verification, is released under the Apache License 2.0. Anyone may use it and change their own copy for any purpose,

keeping the notices and not calling their version xGEO. The game, its art and story, its scenarios (Close Pass among them) and this document stay all rights reserved. Core is open source but not open to contributions: bug reports, physics corrections and feature requests are welcome as issues; pull requests are not accepted. Done: the asset licence audit, the Core and Live split with its check, the command-line and batch runners, the fidelity table, and contribution, conduct and security guides. The jamming and counterspace models are built only from open, published sources, every one cited in this edition, and the owner has cleared them and the name for release. Still to do before release: a review of the repository's history, then making it public. Issues must cite public sources only.

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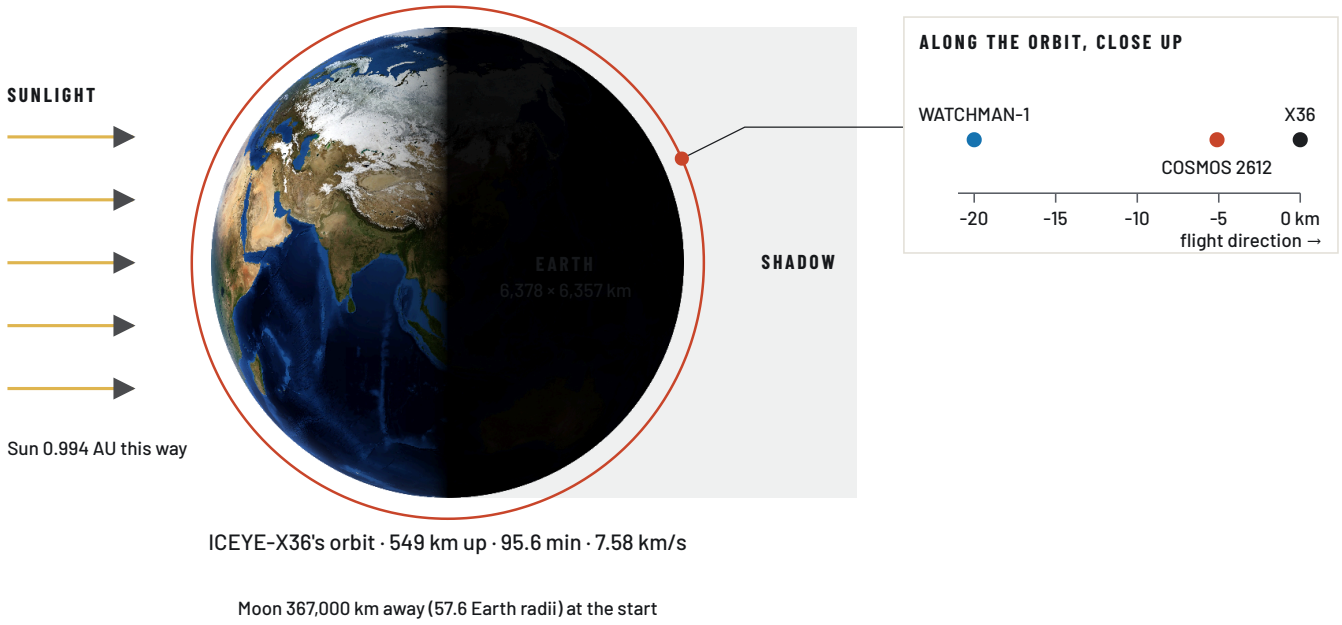
12 WORKED EXAMPLE: CLOSE PASS

This section shows the models of §2–8 working together on one real-world case. Every number here is scenario data or a result of running the engine on it. None of it is a fact of the engine.

12.1 The real event

Close Pass is built on a real uncooperative rendezvous and proximity operation. In October 2026 the Russian satellite Cosmos 2612 approached ICEYE-X36, a commercial X-band radar-imaging satellite, and shadowed it at a few kilometres. It was reported from LeoLabs tracking data by Breaking Defense and The War Zone on 19 October 2026[158,159], and reconstructed from public element sets by KeepTrack[168]. As reported, the sequence was: coplanar alignment; closing from long range; station-keeping within 5 km, then 1–2 km; a closest pass reported at 500 m (another report gives 26 m); and moves ahead of and behind the target. The scenario keeps the real craft and their real orbits. WATCHMAN-1, the US inspector the player commands, is notional, and so is everything Cosmos 2612 does after the start. Blue's task is to make Cosmos 2612 break off without contact, debris or open conflict, while keeping track of it with real ground radars.

FIG. 9 **CLOSE PASS, TO SCALE**



The Earth and ICEYE-X36's orbit drawn to the same scale, with sunlight from the left and the shadow behind. At this scale the three craft sit on one point; the inset spreads them out along the orbit. Start positions from §12.2. Earth imagery: NASA Earth Observatory, Blue Marble Next Generation (Stöckli et al. 2005).

12.2 The craft and where they start

The world is centred on the Earth. The clock starts at 21:00 UTC on 26 October 2026 and runs two days. ICEYE-X36 (NORAD 59103, 2024-043F) starts on its real orbit. Its CelesTrak GP element set (epoch 26 October 2026, 21:06 UTC)[166] was propagated by SGP4[167] and rotated into GCRF (§2.2). The result is a near-circular orbit with semi-major axis 6,924.4 km: 549 km up, a period of 95.57 minutes and a speed of 7.584 km/s.

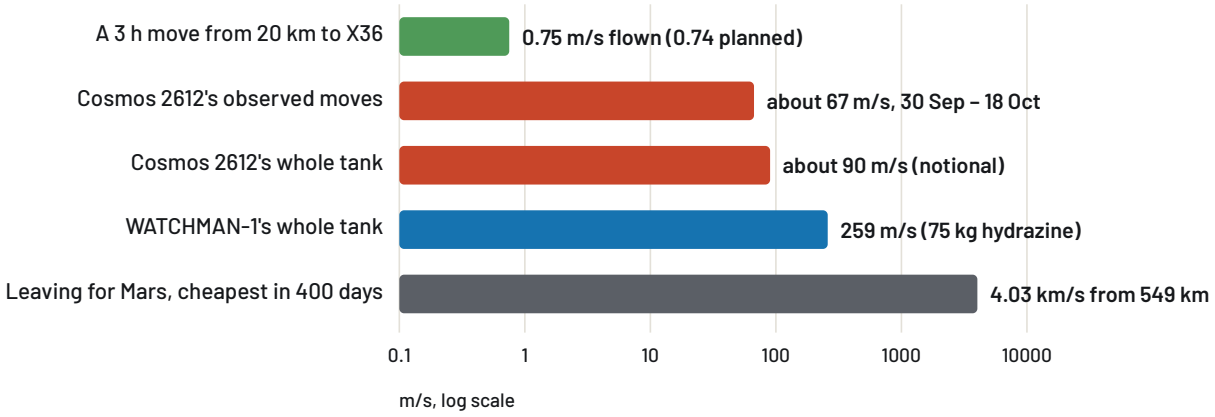
Cosmos 2612 (NORAD 68762, 2026-083H) starts 5.1 km behind ICEYE-X36 and 140 m above it, where public element sets put it at 07:38 UTC on 18 October 2026 (KeepTrack's account of the rendezvous[168]). It is placed on the path that does not drift away along-track. Blue starts with Cosmos 2612's catalogued orbit, about 200 m off its true one (a design value). WATCHMAN-1 starts 20 km behind ICEYE-X36.

CRAFT	ROLE	MASS	ΔV	SENSORS AND RADIO	SOURCE
ICEYE-X36	Target: commercial X-band SAR, operated by ICEYE US	90 kg	none modelled	3.25 m SAR antenna at 9.65 GHz	ESA Earth Online[160]; Gunter's Space Page[169]; GCAT[170]
Cosmos 2612	Red inspector, Russian Aerospace Forces	500 kg	≈ 90 m/s	0.1 m infrared camera; S-band command radio, 5 W into a 0.3 m dish	Mass, engine and radio notional; its observed moves, about 67 m/s from 30 Sep to 18 Oct, from public element sets[168]
WATCHMAN-1	Blue inspector (notional, after GSSAP)	675 kg	259 m/s	0.3 m telescope; 0.5 m receiving dish with a 2 m interferometer baseline	Mass within the 650–700 kg reported for GSSAP[150]; 4 × 22 N hydrazine at 2.2 km/s exhaust speed, 75 kg of propellant: design values

Table 4. The three craft. Values marked notional are DESIGN VALUES; GSSAP's sizes were never published.

Close Pass allows flights between planets, but leaving Earth orbit from 549 km costs at least 4.03 km/s (§3.5). WATCHMAN-1 carries 259 m/s and Cosmos 2612 about 90, so both are held to close-in moves by real physics, not by a rule of the game.

FIG. 10 FUEL AGAINST WHAT THINGS COST



Δv from §3.3, §3.5 and §12.2. Close-in moves are cheap; WATCHMAN-1 can make hundreds of them. Leaving Earth orbit is out of reach of every craft here, which is real physics, not a game limit.

12.3 Five real ground radars

Five real radar sites look for the craft, as §4.5 describes. Range noise is published for three of them; angle noise and masks are notional. PFISR, at 65.1° N, moves 704 km in an hour on the turning Earth.

SITE	OPERATOR	RANGE Σ	ANGLE Σ	MASK	REACH	SOURCE
PFISR, Poker Flat, Alaska	SRI International / NSF; LeoLabs	14 m	0.05°	45°	3,000 km	Site and range noise: Nicolls et al. 2017[171], Table 1
Midland Space Radar, Texas	LeoLabs	14 m	0.1°	30°	3,000 km	Site and range noise: Nicolls et al. 2017[171]
Kiwi Space Radar, New Zealand	LeoLabs	17 m	0.02°	40°	1,500 km	Site, range noise and reach: Stevenson et al. 2020[172]
Arizona Space Radar	LeoLabs	15 m	0.05°	20°	3,000 km	Site: FCC experimental licence WO2XME[173]; noise notional
Space Fence, Kwajalein	U.S. Space Force, 20 SPCS	15 m	0.02°	20°	3,000 km	Notional: not published

Table 5. Ground radar in Close Pass. Angle noise, masks and Space Fence's figures are DESIGN VALUES.

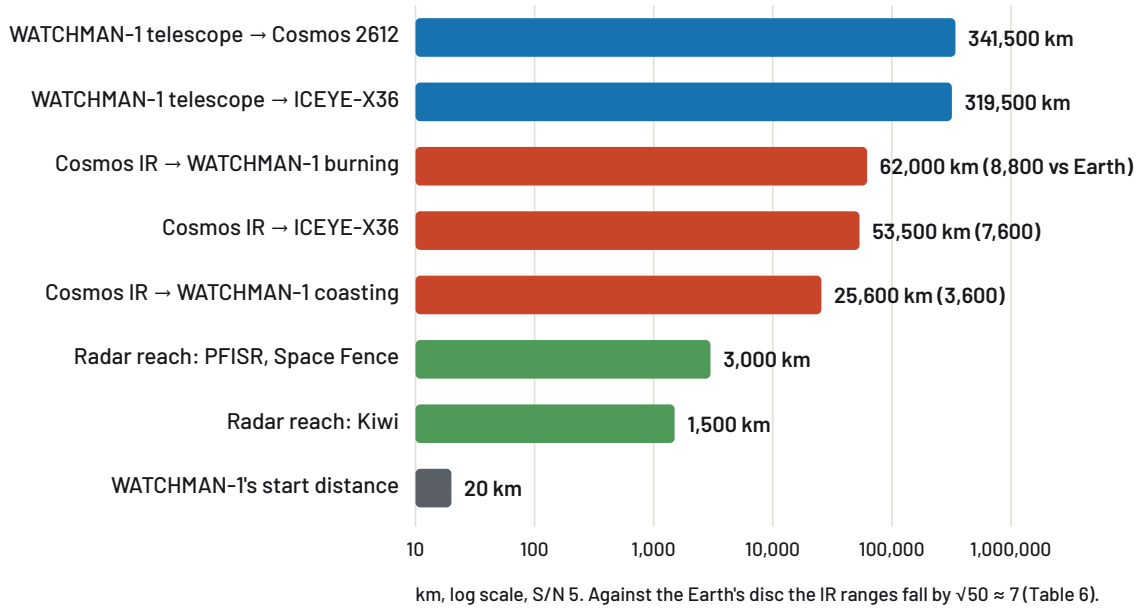
12.4 What the sensors see

Cosmos 2612's command radio aims its 0.3 m dish at its ground station near Moscow, and WATCHMAN-1 listens through a 0.5 m dish of its own, so WATCHMAN-1 hears it only off the main beam (§4.1). WATCHMAN-1's two receivers, 2 m apart, take bearings on what it hears (§4.4). Table 6 gives the ranges at which each telescope detects each craft, from the engine's own functions.

OBSERVER → TARGET	BAND	AGAINST DEEP SKY	AGAINST THE EARTH
WATCHMAN-1's 0.3 m telescope → Cosmos 2612	Visible, sunlit	341,528	—
WATCHMAN-1's telescope → ICEYE-X36	Visible, sunlit	319,470	—
Cosmos 2612's 0.1 m camera → WATCHMAN-1, coasting	6–10 µm	25,611	3,622
Cosmos 2612's camera → WATCHMAN-1, burning	6–10 µm	61,962	8,763
Cosmos 2612's camera → ICEYE-X36	6–10 µm	53,518	7,569

Table 6. Detection ranges at S/N 5, km, at the start ($S = 1{,}361\ W\ m^{-2}$), measured from the engine's own functions. Against the Earth's disc the NEI is ×50, so infrared ranges fall by √50. The telescope looks only where a track already exists (cued), and only at sunlit targets.

FIG. 11 WHO CAN SEE WHOM, AND HOW FAR



From Table 6 and Table 5. Every sensor here outranges the scenario by a wide margin: what limits seeing in Close Pass is geometry (sunlight, the Earth in the way, a radar site's pass), not range.

For the visible ranges the hull is a Lambert sphere (§6.4). With a 1 m telescope the same hulls are seen at 583,270 km (ICEYE-X36), 623,542 km (Cosmos 2612) and 1,150,225 km (WATCHMAN-1), each 1.83 times the 0.3 m range (§6.5). These are long because the Sun at 1 AU is 27 times brighter than at Jupiter.

12.5 Cosmos 2612's plan, and the scenario's design values

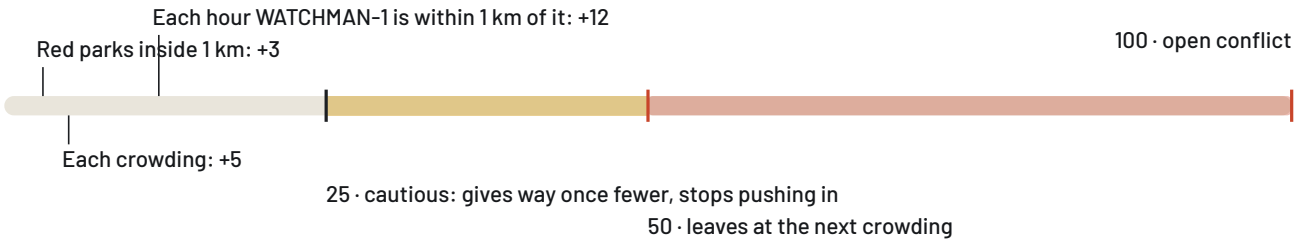
Cosmos 2612 flies the shadow rules of §8.1 with these settings:

- **Its plan.** In the order LeoLabs reported the real one: close in from behind to 2 km, reposition 2 km ahead, then loiter 0.8 km behind. Each move is planned to take 3 hours. The first two slots are held for 6 hours, the last for 8. Distances and times are DESIGN VALUES.
- **Coming closer.** When no Blue craft is within 50 km of it, it takes its next slot at half the distance, but not under 0.5 km.
- **Giving way.** With a Blue craft within 5 km of it, it gives way to a slot 8 km on the far side of ICEYE-X36. Crowded again after two give-ways (one in the tutorial), it leaves: 4 m/s prograde, and gone once it is beyond 200 km.
- **Its radio.** An S-band (2.25 GHz) BPSK burst at 100 kbaud, 5 W into a 0.3 m dish aimed at its ground station near Moscow (the Titov control centre is at Krasnoznamensk), two minutes long, every 3 to 6 hours and after each move. These are design values within ordinary practice.

- **Losing its link.** Cut off from its ground station for 3 hours within 200 km of its target, and with a Blue craft within 25 km of it, it leaves. Cut off but alone, it holds its slot and waits. The threshold and the burn are DESIGN VALUES.
- **Score.** Making it break off counts 40; keeping it outside 1 km of ICEYE-X36, 20; custody (the track's 1σ under 1 km, and the track truly within 1 km, at least 80% of the time), 20; using no more than half of WATCHMAN-1's fuel, 10; staying below open conflict, 10. Design values.
- **Timescale.** The real event took weeks. The scenario runs two days from 21:00 UTC on 26 October 2026 DESIGN VALUE.

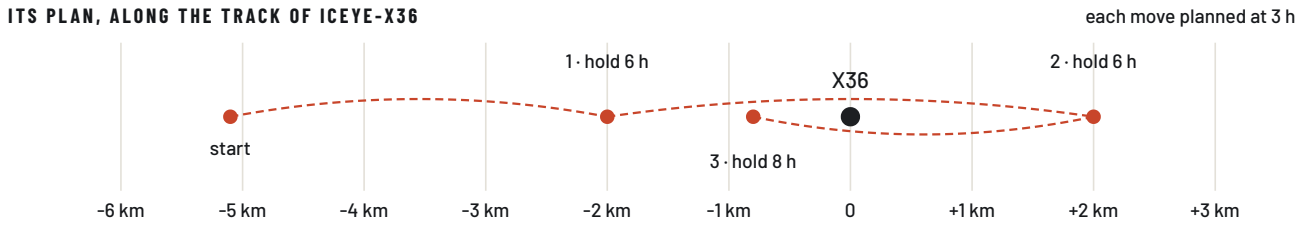
Blue has no effects in Close Pass. WATCHMAN-1's only lever is its presence. The scenario's tension rules (§5.3) price that presence. Tension rises by 3 when Cosmos 2612 parks inside ICEYE-X36's 1 km keep-out, by 5 each time WATCHMAN-1 crowds it into giving way, and by 12 for every hour WATCHMAN-1 spends within 1 km of it. At 25, Cosmos 2612 turns cautious: it gives way once fewer before leaving, and it stops pushing in. At 50 it leaves at the next crowding. At 100 the confrontation has become open conflict and the scenario is lost. Every value is a DESIGN VALUE.

FIG. 12 TENSION IN CLOSE PASS



The tension rules of §12.5. Every value is a DESIGN VALUE. Doctrine gives no ladder (§8.2); this one rewards pressure that stops short of contact.

FIG. 13 **HOW COSMOS 2612 BEHAVES**



WHAT CHANGES ITS PLAN

No Blue craft within 50 km	takes its next slot at half the distance (not under 0.5 km)
Blue within 5 km of it	gives way to a slot 8 km on the far side of X36
Crowded again after 2 give-ways	leaves: +4 m/s prograde, gone beyond 200 km
Cut off from its handler 3 h, Blue within 25 km	leaves; cut off but alone, it waits

Top: its plan along ICEYE-X36's track (§12.5). Bottom: what makes it come closer, give way or leave. Every threshold is a DESIGN VALUE.

12.6 A day left alone

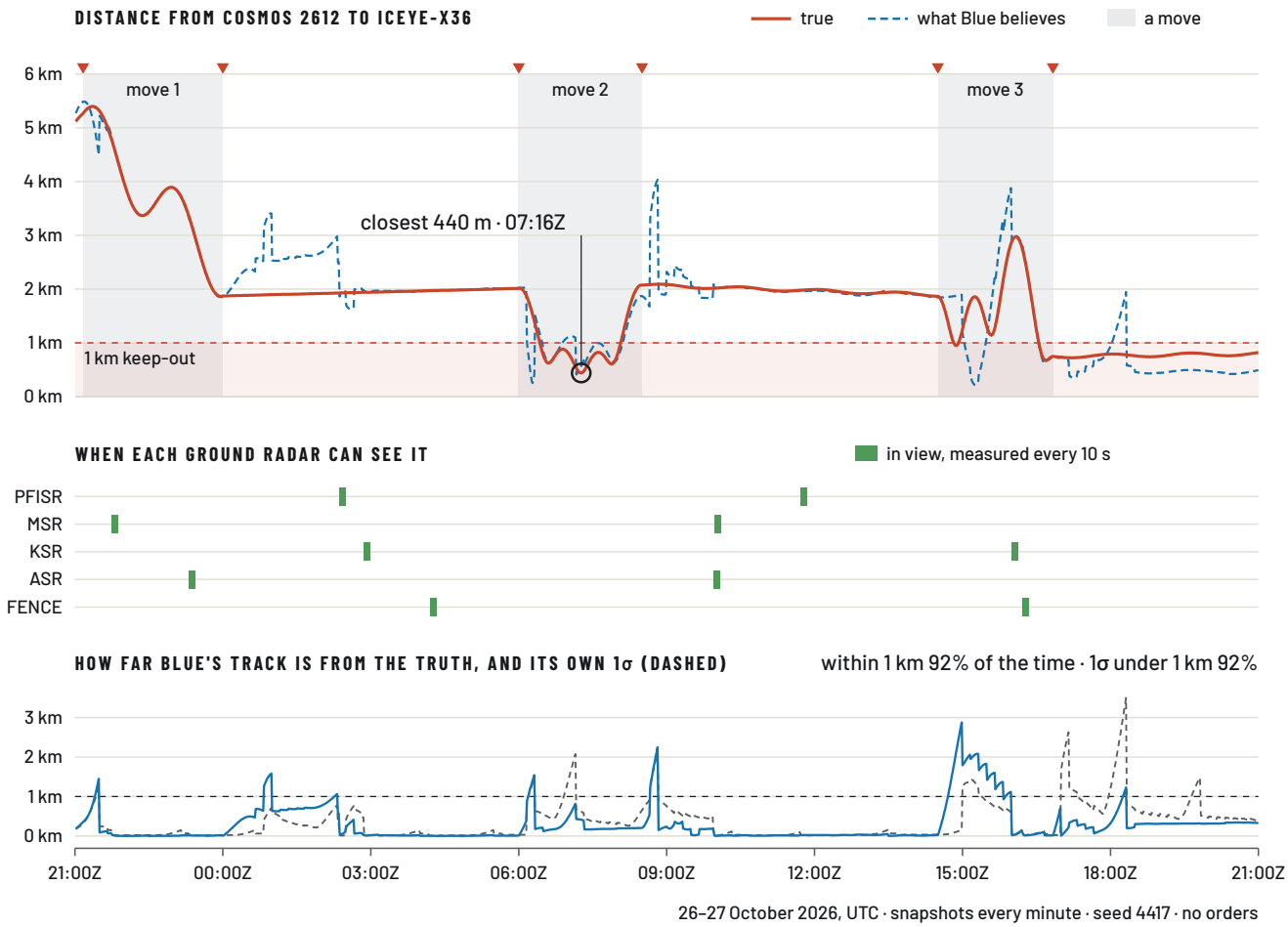
Fig. 14 is a real 24-hour run of the engine on Close Pass with no orders given. It was made with `cli/run.gd --hours=24 --step=60 --truth=true`, seed 4417, so it has one snapshot a minute and the true states beside Blue's belief.

Cosmos 2612 works through its plan in three moves, each a burn to leave and a burn to stop. It parks 2 km behind ICEYE-X36 at midnight. Between 06:00Z and 08:30Z it crosses to its slot 2 km ahead, and on the way it passes 440 m from ICEYE-X36, at 07:16Z. From 14:30Z it comes back to 0.8 km behind, inside the keep-out, and stays there. In all it spends 6.2 hours inside 1 km, so the keep-out objective fails. WATCHMAN-1, given no orders, drifts from 20 km to 41 km behind.

The ground radars see Cosmos 2612 in ten short windows, two per site, each 2 to 5 minutes long, 30 minutes in all; each site measures it every 10 s through each window. Between those, WATCHMAN-1's telescope takes bearings and ranges it by its size whenever Cosmos 2612 is in sunlight, and the filter predicts across each pass through the Earth's shadow. Blue's track was within 1 km of the truth 92% of the time, and its own 1σ was under 1 km 92% of the time: the track is about as sure as it is right. Its worst error was 2.9 km, at 14:59Z, after a burn made while no sensor could see it.

Two numbers in the run's own debrief differ from the figure, and both differences are expected. The debrief checks at each 10-minute look, so it reports the closest approach as 490 m, where the one-minute snapshots find 440 m; and its custody, which needs both the 1σ and the true error under 1 km (§7), is 90%, where the figure counts each alone.

FIG. 14 CLOSE PASS LEFT ALONE FOR 24 HOURS



A real Core run of Close Pass with no orders: `cli/run.gd --hours=24 --step=60 --truth=true`, seed 4417, one snapshot a minute. Top: the distance from Cosmos 2612 to ICEYE-X36, true and as Blue's track puts it. The shaded bands are its three moves; each triangle is a burn, found from its fuel use. Middle: when each ground radar could see it, above its mask and within reach; each site measures every 10 s of each window. Bottom: the distance between Blue's track and the truth (solid), and Blue's own 1σ (dashed), both at the snapshot's own time.

12.7 What is notional, and what is not known

- Cosmos 2612's mass, engine, propellant, camera and radio are notional. Only its orbit and its observed moves are public.
- WATCHMAN-1 as a whole is notional. GSSAP's mass is reported; its sizes, thrusters and sensors were never published.
- The radar sites' angle noise and elevation masks, and all of Space Fence's figures, are notional.
- The site and mask of Cosmos 2612's ground station are notional.

- The closest approach of the real event is not settled: one report gives 500 m, another 26 m.
- Drag is not modelled (§10). Over two days it would move the three craft relative to each other by an amount that depends on their area-to-mass ratios, which are not public.

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